

EXHAUSTIVE ACADEMIC REVIEW PROTOCOL

ELECTRICAL INSTRUMENTATION: PHASE 5

The Edge Cases & Transient Numericals

Microphone Dynamics, Piezoelectric Decay, Conducting Fluids, and Diaphragms

Semester Preparation Masterclass

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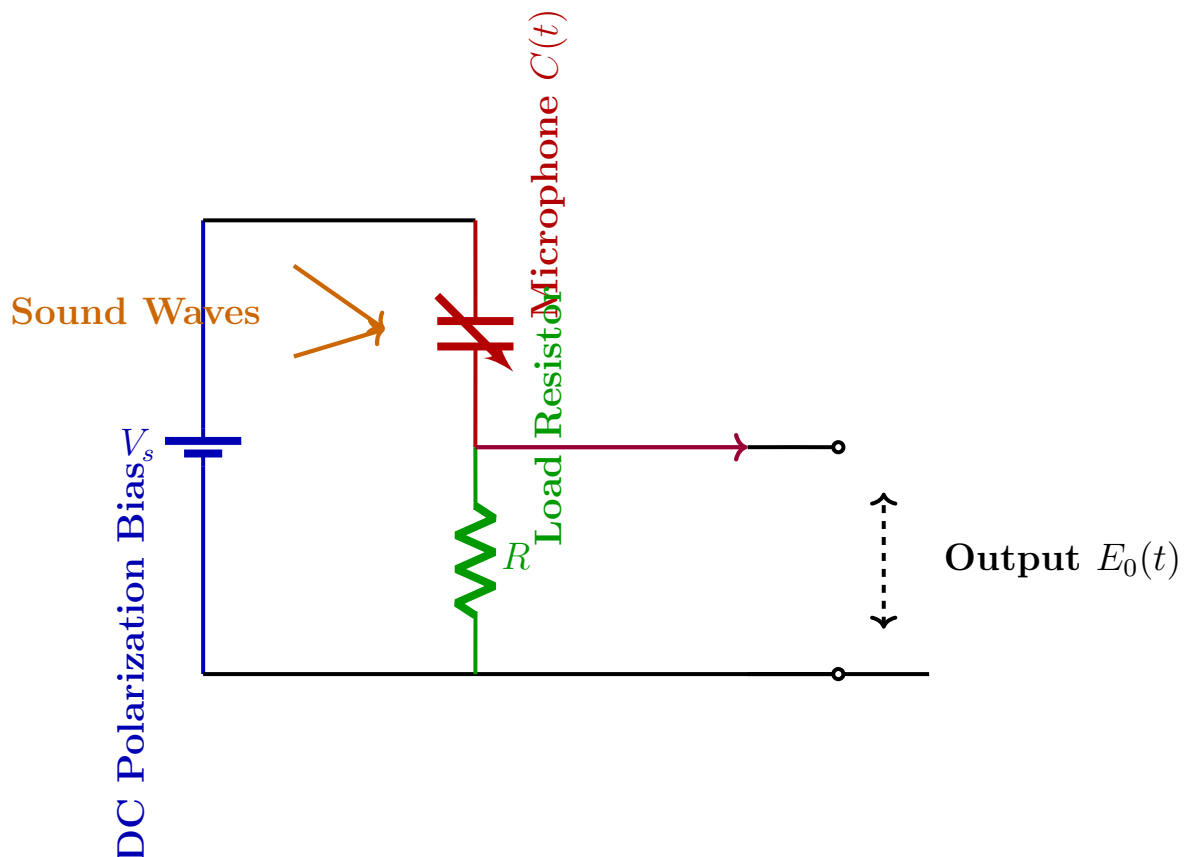
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The Capacitor Microphone (Dynamic Acoustics)

The Capacitor Microphone is a specialized variable-gap capacitive transducer utilized for detecting high-frequency dynamic acoustic pressure waves (sound). It differs from a standard capacitive displacement sensor because it requires a high-voltage DC bias (V_s) and operates strictly in the dynamic domain via a series load resistor (R).

Physical Architecture

The physical assembly consists of a massive, rigid metallic backplate and a highly flexible, ultra-thin metallic diaphragm stretched parallel to it. When acoustic waves strike the diaphragm, it dynamically deflects, causing the gap distance d to change over time, resulting in a time-varying capacitance $C(t)$.



Grassroots Derivation of the Transfer Function

By applying Kirchhoff's Voltage Law (KVL) around the single closed series loop, we sum the voltage drops across the resistor and the dynamic capacitor:

$$V_s = V_R + V_C = i(t)R + \frac{1}{C(t)} \int i(t)dt \quad (1)$$

Because the DC source V_s is a rigid constant, taking the mathematical time derivative of the entire equation yields zero on the left side:

$$0 = R \frac{di(t)}{dt} + \frac{i(t)}{C(t)} + \left[\int i(t)dt \right] \frac{d}{dt} \left(\frac{1}{C(t)} \right) \quad (2)$$

Let the gap distance at rest be d_0 . The dynamic displacement caused by the sound wave is $x_1(t)$. The physical distance becomes $d(t) = d_0 - x_1(t)$. The capacitance becomes:

$$C(t) = \frac{\epsilon A}{d_0 - x_1(t)} = \frac{\epsilon A}{d_0(1 - x_1/d_0)} \approx C_0 \left(1 + \frac{x_1(t)}{d_0} \right) \quad (3)$$

where $C_0 = \frac{\epsilon A}{d_0}$ is the static resting capacitance.

In a grassroots perturbation analysis, assuming the dynamic displacement $x_1(t)$ is microscopically small compared to the resting gap d_0 , the charge on the capacitor $q(t)$ remains approximately constant at its static bias level $Q_0 = V_s C_0$. Therefore, the current $i(t) = \frac{dq(t)}{dt}$ is governed entirely by the geometric change in capacitance.

The output voltage is extracted across the resistor: $e_0(t) = i(t)R$. By substituting these small-signal approximations into the differential equation and taking the Laplace transform, the complex differential equation collapses into a standard first-order **High-Pass Filter** transfer function:

$$\frac{E_0(s)}{X_1(s)} = \left(\frac{V_s}{d_0} \right) \left[\frac{s\tau}{1 + s\tau} \right] \quad (4)$$

where the critical electrical time constant is defined as $\tau = R \cdot C_0$.

Numerical Blueprint: Cut-off Frequency & Resistance

BJB Sir frequently tests the mathematical limits of this high-pass filter. Substituting $s = j\omega$, the frequency response magnitude is:

$$|G(j\omega)| = \left(\frac{V_s}{d_0}\right) \frac{\omega\tau}{\sqrt{1 + (\omega\tau)^2}} \quad (5)$$

The Trap: The question will ask you to calculate the required resistance R to achieve a specific low-frequency cut-off (f_c) with a specified amplitude tolerance (e.g., 5% error allowed). 1. A 5% error means the magnitude must be 0.95 of its maximum flat-band value.

$$\frac{\omega\tau}{\sqrt{1 + (\omega\tau)^2}} = 0.95 \quad (6)$$

2. Square both sides to eliminate the radical:

$$\frac{(\omega\tau)^2}{1 + (\omega\tau)^2} = 0.9025 \implies (\omega\tau)^2 = 0.9025 + 0.9025(\omega\tau)^2 \quad (7)$$

$$0.0975(\omega\tau)^2 = 0.9025 \implies \omega\tau = \sqrt{\frac{0.9025}{0.0975}} \approx 3.04 \quad (8)$$

3. Since $\omega = 2\pi f_c$, you can isolate the required time constant τ :

$$\tau = \frac{3.04}{2\pi f_c} \quad (9)$$

4. Finally, since $\tau = R \cdot C_0$, you divide by the base capacitance to find the massive required resistance:

$$R = \frac{\tau}{C_0} \quad (10)$$

(Note: Because C_0 is usually in picofarads, R will mathematically demand values in the tens or hundreds of Mega-Ohms).

The Piezoelectric Transient Numerical

In the 2025 examination, BJB Sir bypassed standard theory and deployed a brutal mathematical transient-decay numerical. You must execute this exact step-by-step framework to survive it.

The Problem Architecture

Typical Problem Statement: A piezoelectric sensor is connected to a voltmeter via a 20m coaxial cable. A sudden static step-force of 10 N is applied. Calculate the exact voltage indicated by the voltmeter $t = 2$ seconds later.

The Electrical Framework: The crystal, the cable, and the voltmeter all exist in a strictly **parallel** electrical topology. You must first calculate the lumped equivalent parameters.

Step 1: Total Equivalent Capacitance (C_{eq}) Capacitors in parallel add directly.

$$C_{eq} = C_{sensor} + C_{cable} + C_{voltmeter} \quad (11)$$

(Trap: Cable capacitance is usually given per meter, e.g., 40 pF/m. You must multiply this by the length of the cable!)

Step 2: Total Equivalent Resistance (R_{eq}) Resistors in parallel must be combined using the product-over-sum rule. The crystal has a massive leakage resistance (R_s), and the voltmeter has an input impedance (R_v).

$$R_{eq} = \frac{R_s \cdot R_v}{R_s + R_v} \quad (12)$$

Step 3: The System Time Constant (τ) Multiply the lumped equivalents (ensure units are strictly in Ohms and Farads to yield Seconds).

$$\tau = R_{eq} \cdot C_{eq} \quad (13)$$

Step 4: Initial Peak Voltage (V_{peak}) When the step force (F_0) strikes, the generated

charge q is instantaneous.

$$q = d \cdot F_0 \quad (d \text{ is the charge sensitivity constant, C/N}) \quad (14)$$

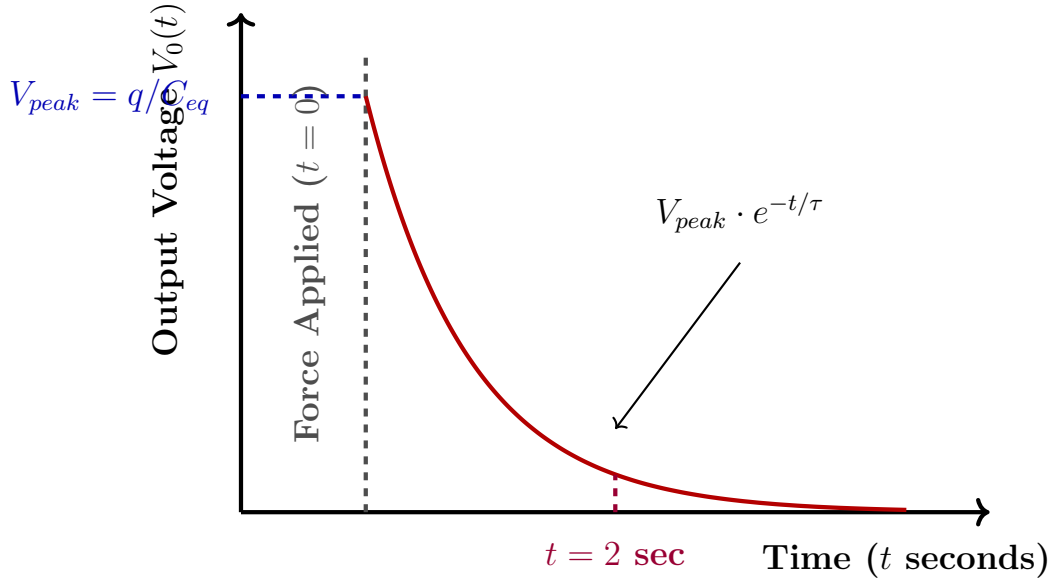
The initial peak voltage developed across the parallel network at exactly $t = 0^+$ is:

$$V_{peak} = \frac{q}{C_{eq}} \quad (15)$$

Step 5: The Exponential Decay Because a static force produces no continuous dynamic current ($s = 0$), the initial charge violently bleeds away to ground through the equivalent resistance R_{eq} . The output voltage undergoes a pure exponential decay governed by τ :

$$V_0(t) = V_{peak} \cdot e^{-t/\tau} \quad (16)$$

Simply substitute the requested time t into the exponent and calculate the final microvolt reading.



Conducting Liquid Level Gauge

In Phase 1, we derived the standard capacitive level gauge for **non-conducting** (dielectric) fluids like oil. However, BJB Sir frequently traps students by specifically asking for the **Conducting Liquid** variant (e.g., water, acids).

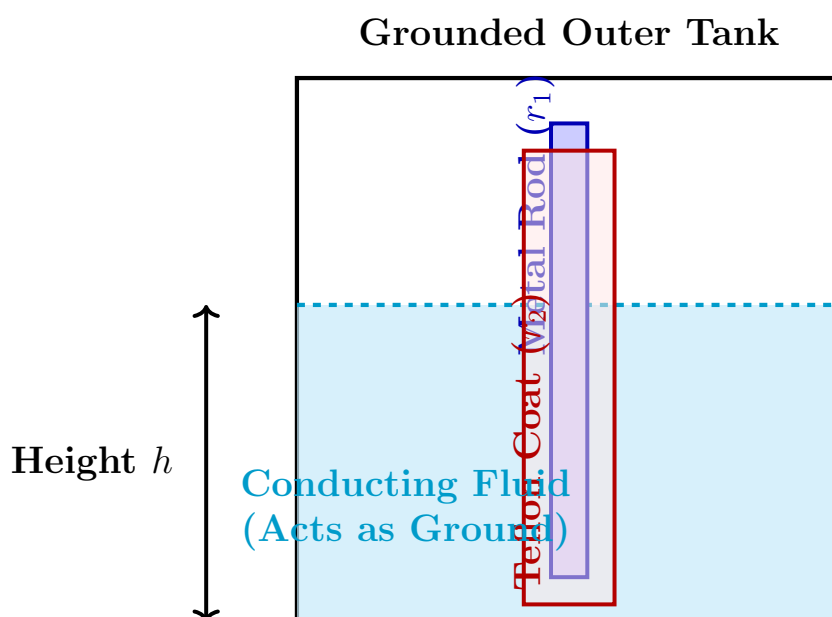
The Engineering Problem

If the liquid is electrically conductive, you cannot use a bare metallic inner rod. The conductive liquid would instantly short-circuit the inner rod to the grounded outer tank wall, collapsing the capacitance to zero and destroying the power supply.

The Teflon Fix & Architecture

To measure a conductive liquid, the central metallic rod **MUST** be heavily coated in a rugged, chemically inert dielectric material, universally **Teflon (PTFE)**.

In this topology, the conductive liquid itself acts as the **Outer Electrode** (electrically continuous with the grounded tank). The Teflon coating acts as the sole dielectric medium. The metallic core of the rod acts as the Inner Electrode.



Mathematical Derivation

Let r_1 be the absolute radius of the inner metallic rod. Let r_2 be the outer radius of the Teflon coating. Let ϵ_{tef} be the relative permittivity of the Teflon layer.

Because the liquid is a conductor, the electric field only exists across the Teflon layer. Furthermore, the capacitor only exists exactly where the conductive liquid physically embraces the Teflon-coated rod (height h). The air/vapor above the liquid does not contribute to the measurement because the outer "plate" (the liquid) is missing.

Therefore, the system is a single, simple coaxial capacitor of variable length h :

$$C_{total} = \frac{2\pi\epsilon_0\epsilon_{tef}}{\ln(r_2/r_1)} \cdot h \quad (17)$$

The Sensitivity (S) is the mathematical derivative with respect to height h :

$$S = \frac{dC_{total}}{dh} = \frac{2\pi\epsilon_0\epsilon_{tef}}{\ln(r_2/r_1)} \quad (18)$$

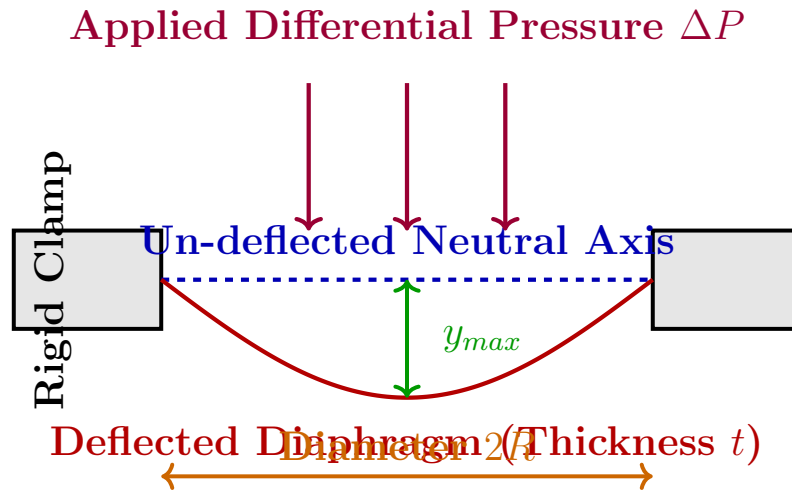
Conclusion: For a conducting fluid, the sensitivity is completely independent of the fluid's own permittivity. It is strictly dictated by the geometry and permittivity of the Teflon jacket. It yields a perfectly linear response.

Diaphragm Pressure Transducers

The diaphragm is the supreme primary mechanical transducer for translating fluid or gas pressure into physical displacement. It is universally tested in BJB Sir's exams due to its mechanical-to-electrical signal conversion requirements.

Physical Mechanics & Deflection

A diaphragm is a highly elastic, thin circular metallic or silicon plate. Its entire outer circumference is rigidly clamped by the sensor housing. When a differential pressure ($\Delta P = P_1 - P_2$) is applied across its faces, the diaphragm bends and deflects.



Using the theory of flat elastic plates, the maximum deflection (y_{max}) occurs at the exact geometric center ($r = 0$) and is governed by:

$$y_{max} = \frac{3(1 - \nu^2)R^4}{16Et^3} \cdot \Delta P \quad (19)$$

Where:

- E is Young's Modulus of the material (stiffness).
- ν is Poisson's Ratio.
- R is the unclamped radius, and t is the material thickness.

The Linearity Constraint: This equation is strictly a *linear approximation* that as-

sumes membrane stretching stresses are negligible compared to bending stresses. For the sensor to remain mathematically linear, engineering law dictates the deflection must not exceed one-third of the thickness:

$$y_{max} \leq \frac{t}{3} \quad (20)$$

If pressures force the deflection beyond this absolute limit, the diaphragm violently stiffens, introducing massive non-linear cubic errors ($\Delta P = Ay_{max} + By_{max}^3$). To linearize massive pressure ranges, engineers replace flat plates with **Corrugated Diaphragms** (concentric ripples) to artificially extend the elastic stretch zone.

Secondary Electrical Conversion

A mechanical deflection is useless to a microcontroller. It must be converted to an electrical signal via a secondary transducer.

Method A: Capacitive Pickup The metallic diaphragm acts as the moving plate of a variable air-gap capacitor, with a rigid fixed electrode placed at a microscopic distance d below it. Because this introduces the hyperbolic $1/d$ non-linearity discussed in Phase 1, it must be paired with a differential Push-Pull arrangement and an AC Bridge to linearize the pressure readout.

Method B: Piezoresistive Strain Gauges When the diaphragm bends, it generates maximum tangential stress at the absolute center, and maximum radial stress at the clamped outer edges. Four semiconductor strain gauges are bonded using epoxy to these precise high-stress locations. As the diaphragm deflects, the gauges stretch and compress, altering their electrical resistance via the piezoresistive effect. These four resistors form a full **Wheatstone Bridge**, converting the pressure directly into a calibrated millivolt DC output.